Smart Infra & Automation

Project Name: Automated Waste Segregation





The Problem: Aarav notices that students in his school throw wet and dry waste into the same dustbin. This makes it difficult to separate the waste and recycle it properly. He thinks there should be a smart dustbin that can identify the type of waste by itself. Aarav decides to build a system that can check the waste and move it to the correct bin.

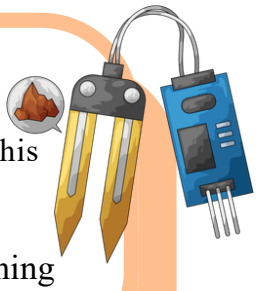
🎯 **What You Will Design:** Students will build a smart waste segregation system that can identify wet and dry waste using sensors. The system will automatically move the waste into the correct dustbin and show the result on the OLED display.



🌱 What is a Soil Sensor?

A Soil Sensor can detect the amount of moisture or water present in an object. In this project, it helps us identify whether the waste is wet or dry.

Example: 🌧️ Just like you can tell whether the ground is wet after rain by touching it, the soil sensor can detect moisture electronically.



Components Required









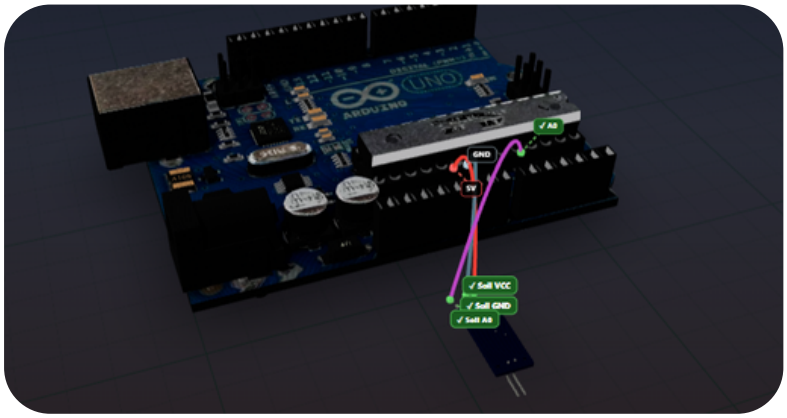




Circuit Designing

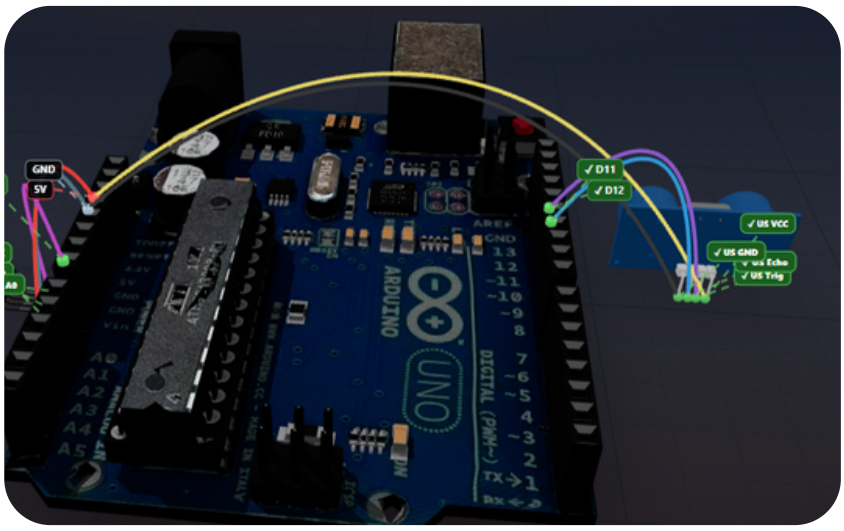
Step 1: Connect soil sensor to the Arduino Uno.

Sensor Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
A0 Pin	A0 Pin



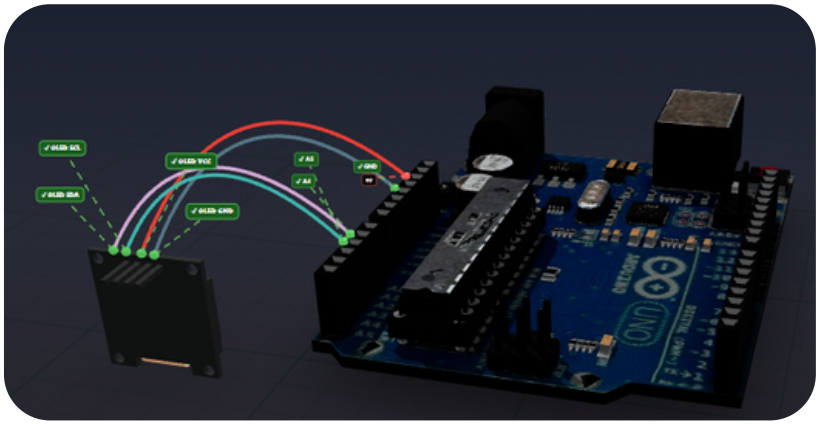
Step 2: Connect the ultrasonic sensor to the Arduino Uno.

Sensor Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
Trig Pin	Digital Pin 12
Echo Pin	Digital Pin 11



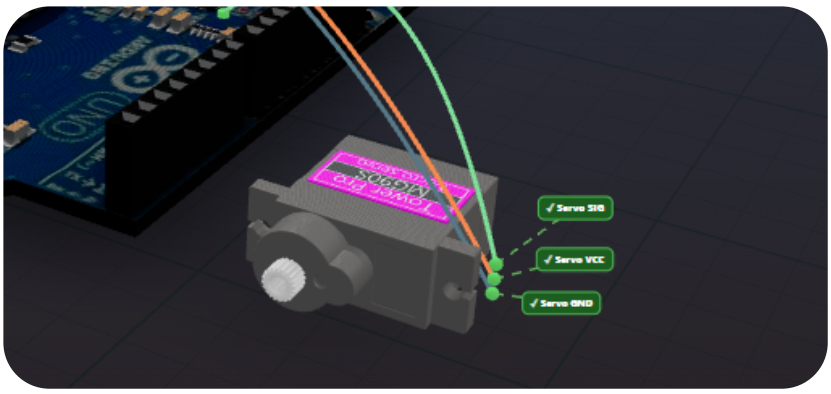
3. Connect the OLED display to the Arduino Uno.

OLED Pins	Arduino Pins
Vcc Pin	5V Pin
GND PiN	GND Pin
SCL Pin	Analog A5 Pin
SDA Pin	Analog A4 Pin



4. Connect the servo motor to the Arduino Uno.

Motor Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
Signal Pin	Digital Pin 8



Project Coding

1. Open the Arduino Uno software and write code to add the libraries and define the pins.

```
#include <Servo.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

Servo servo1;
```

2. Make code to define the sensor pins and define the variables.

```
// Ultrasonic Sensor Pins
const int trigPin = 12;
const int echoPin = 11;

// Soil Sensor Pin
const int soilPin = A0;

// Variables
long duration;
int distance = 0;
int soil = 0;
int fsoil = 0;

// Threshold Values
int maxDryValue = 30;       // Moisture threshold
int Ultra_Distance = 15;    // Detection distance in cm
```

3. Make code to set up the OLED display screen and define the trig and echo pins of the ultrasonic sensor.

```
30  void setup()
31  {
32    Serial.begin(9600);
33    pinMode(trigPin, OUTPUT);
34    pinMode(echoPin, INPUT);
35    servo1.attach(8);
36    servo1.write(90);
37    // OLED Setup
38    if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
39    {
40      while (true);
41    }
42    display.clearDisplay();
43    display.setTextColor(SSD1306_WHITE);
```

4. Make code to display the welcome message on the OLED display.

```
45    // Welcome Screen
46    display.setTextSize(2);
47    display.setCursor(5, 10);
48    display.println("SMART");
49    display.setCursor(5, 35);
50    display.println("WASTE BIN");
51    display.display();
52
53    delay(3000);
54
55    readyScreen();
56  }
```

5. Make code to calculate the distance between the ultrasonic sensor and the waste and display the values on the OLED screen.

```
58  void loop()
59  {
60    distance = getDistance();
61    if (distance < Ultra_Distance && distance > 1)
62    {
63      display.clearDisplay();
64      display.setTextSize(1);
65      display.setCursor(10, 20);
66      display.println("Checking Waste...");
67      display.display();
68      delay(1000);
69      fsoil = getMoisture();
70      Serial.print("Moisture = ");
71      Serial.print(fsoil);
72      Serial.println("%");
```

6. Make code for the wet waste and display the wet waste message on the OLED display.

```
74      // Wet Waste
75      if (fsoil > maxDryValue)
76      {
77        Serial.println("WET WASTE");
78        display.clearDisplay();
79        display.setTextSize(2);
80        display.setCursor(15, 15);
81        display.println("WET");
82        display.setCursor(5, 40);
83        display.println("WASTE");
84        display.display();
85        servo1.write(170);
86        delay(3000);
87      }
```

7. Make code for the dry waste and display the dry waste message on the OLED display.

```
89      // Dry Waste
90      else
91      {
92        Serial.println("DRY WASTE");
93        display.clearDisplay();
94        display.setTextSize(2);
95        display.setCursor(15, 15);
96        display.println("DRY");
97        display.setCursor(5, 40);
98        display.println("WASTE");
99        display.display();
100       servo1.write(10);
101       delay(3000);
102     }
```

8. Make code to return the servo motor to the center position.

```
104       // Return Servo to Center
105       servo1.write(90);
106       delay(1000)
107       readyScreen();
108     }
109     delay(200);
110   }
```

9. Make code to calculate the total distance for the ultrasonic sensor.

```
int getDistance()
{
  long totalDistance = 0;
  for (int i = 0; i < 3; i++)
  {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    duration = pulseIn(echoPin, HIGH);
    int d = duration * 0.034 / 2;
    totalDistance += d;
    delay(10);
  }
  return totalDistance / 3;
}
```

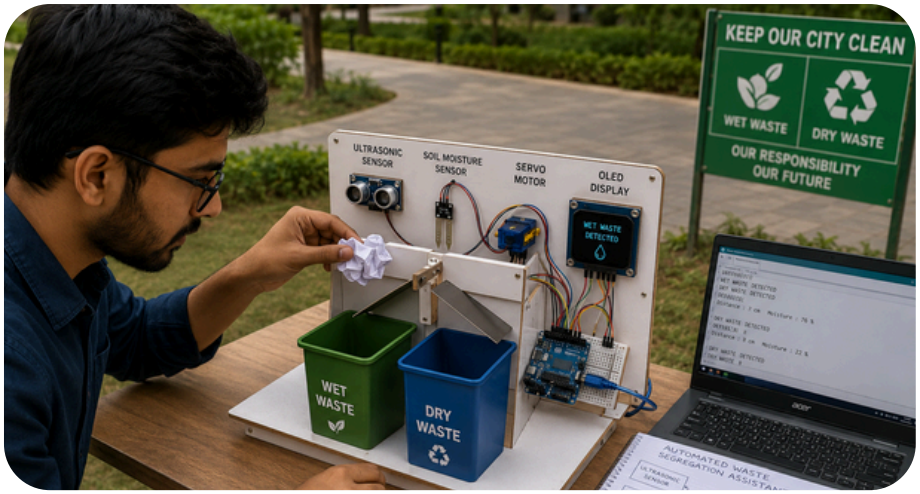
10. Make code to get the reading from the soil sensor.

```
// Function: Soil Sensor
//---------------------------
int getMoisture()
{
  int total = 0;
  for (int i = 0; i < 5; i++)
  {
    soil = analogRead(soilPin);
    soil = constrain(soil, 485, 1023);
    int moisture = map(soil, 485, 1023, 100, 0);
    total += moisture;
    delay(50);
  }
  return total / 5;
}
```

11. Make code to display all the reading values on the OLED display screen.

```
// OLED Ready Screen
//----------------------------
void readyScreen()
{
  display.clearDisplay();
  display.setTextSize(1);
  display.setCursor(15, 10);
  display.println("Automated Waste");
  display.setCursor(30, 25);
  display.println("Segregator");
  display.setCursor(10, 50);
  display.println("Place Waste Here");
  display.display();
}
```

Project Output





🌾 Where Is This Used in Real Life?

This type of smart waste segregation system is used in smart cities, recycling centers, and large waste management plants to separate wet and dry waste automatically.

Learning Outcomes

- The students will be able to connect and use Ultrasonic and Soil Moisture Sensors in an Arduino project.
- The students will be able to write code that reads sensor data and makes decisions based on conditions.
- The students will be able to build an automatic waste sorting system using sensors, a servo motor, and an OLED display.



Section 1: Multiple Choice Questions

1. What problem does Aarav want to solve?

a) Students are not bringing lunch
b) Wet and dry waste are mixed together
c) Dustbins are too small
d) The school needs more chairs

2. Which sensor helps the system identify whether waste is wet or dry?

a) Ultrasonic Sensor
b) Soil Moisture Sensor
c) Light Sensor
d) Touch Sensor

3. What does the servo motor do after the waste is identified?

a) Measures the waste
b) Detects moisture
c) Moves the waste toward the correct dustbin
d) Displays the waste type

Section 2: True and False Statements

1. Mixing wet and dry waste can make recycling difficult.....

2. The ultrasonic sensor helps detect the presence of waste.....

3. The servo motor is used to store the waste inside the dustbin.....